

Utsaw Kumar

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Education

Year(s)	Degree/Certificate	Institute	CPI
2025 - 2027	M.Tech, Computer Science & Engg.	Indian Institute of Technology Bhubaneswar	8.00/10
2021 - 2025	B.Tech, Computer Science & Engg.	Mahatma Gandhi Central University, Motihari	8.88/10
2021	Senior Secondary (XII)	M.S. College, Motihari	85.8%
2019	Matriculation(X)	Arya Vidyapith, Motihari	90.2%

Experience

Freelance Android Developer

2023 – 2025

Self-Employed — Java/Kotlin, Android Studio

- Build and delivered 10+ native Android applications end-to-end, which includes notification system, breathing exercise and Technical-calculator apps, based on client requirements, maintaining a 5.0/5 rating across all reviews.

Projects

Minimum cost Fault-Tolerant Multi-Agent LLM Systems — MTP, IIT Bhubaneswar

Aug 2026 – Ongoing

Supervisor: Dr. Joy Chandra Mukherjee

- Formulated a new graph optimization problem generalizing the Steiner Network problem to design cost efficient multi-agent LLM topologies which guarantee task completion even with failure of k agents.
- Modeled a dynamic, history dependent token costs and formalized the objective as minimizing expected topology cost subject to vertex disjoint connectivity between critical agents.
- Surveyed fault-tolerant MAS benchmarking, cost-aware topology synthesis, and reliability theory to identify the gap, and built an interactive web-based visualizer to demonstrate the problem.

Decentralized P2P Chat Application

Jan 2026 – Feb 2026

C++, TCP/IP, Socket Programming, Non-Blocking I/O

- Developed a decentralized peer-to-peer chat application where each peer can act as both a client and server and establish multiple TCP connections.
- Implemented non-blocking I/O to handle concurrent peer connections within a single-threaded architecture.
- Built private and broadcast messaging, peer discovery and management, connection establishment, and connection termination.

FIFO and Critical-Path Aware Scheduling in SimpleScalar

Aug 2025 – Sep 2025

Advanced Computer Architecture Project — C, SimpleScalar Simulator

- Modified the SimpleScalar `sim-outorder` scheduler module and redesigned RUU ready-queue handling to implement FIFO and critical-path-aware instruction scheduling.
- Benchmarked in 5 SPEC-style workloads measuring `sim_cycle`, IPC and latency against baseline.
- Reduced average RUU latency by 0.5% and 0.46% IPC improvement on I/O-heavy workloads.

AI-Powered Media Transformation SaaS

Sept 2026 - Ongoing

Next.js, TypeScript, Cloudinary, Clerk

- Building a web based SaaS platform where authenticated users can upload, transform, and optimize images or videos using Cloudinary's AI-driven processing.
- Implemented secure user authentication and route protection using Clerk middleware.

Technical Skills

Languages: C++, C, Python, TypeScript, SQL

Networking & Systems: TCP/IP, Socket Programming, Non-Blocking I/O, Concurrency

Core Computer Science: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Computer Architecture

Web Development: Next.js, React, REST APIs

Tools: Git, Linux, Android Studio

Problem Solving

- Solved 350+ DSA problems across LeetCode and GeeksforGeeks, covering trees, graphs, binary search, dynamic programming, and complexity analysis using C++.

Position of Responsibility

Teaching Assistant — Operating Systems Lab

- Guided students through system programming exercises and debugging, providing feedback on lab assignments and problem-solving approaches.